GEOSPATIAL RAP TOOL

User Manual

Prevalence mapping and spatial-cluster analysis

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Language	English
Status	Aligned with the supplied current application source

Purpose. This document describes the current workflow and implementation represented by the supplied project files. It replaces older instructions that referred to previous navigation, storage, or output structures.

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1. Purpose and scope

The Geospatial RAP Tool aggregates cases and population for a selected country, administrative level, year, and epidemiological week window. It produces prevalence maps, a Global Moran's I summary, and Local Moran's I (LISA) cluster classifications.

All units are analysed together. The standalone application does not require selecting each commune, department, or other location separately. After a geographic level is chosen, all matching units available in the active data are processed in the same run.

Interpretation. Spatial statistics describe patterns in the selected dataset. They do not establish causation, identify a source of transmission, or replace local epidemiological investigation.

2. Input data and spatial requirements

Upload one or more API-POP-compatible CSV files. Comma- and semicolon-delimited files are supported. Population is required because prevalence is calculated as cases divided by population times the configured rate multiplier, currently 100,000.

Internal column	Required	Meaning	Accepted aliases or examples
year	Yes	Epidemiological year.	year, anio, año, epiyear
week	Yes	Epidemiological week.	week, semana, epiweek, se
country	Recommended	Country or national unit.	Argentina or England
level1	Depends on selected level	First administrative level.	province, region, Level1
level2	Required for second-level maps	Second administrative level.	department, district, LAD, Level2
level3 / level4	Required for detailed maps	More detailed administrative levels.	fraction, MSOA, Level3
n_cases	Yes	Observed cases or event count.	n_cases, cases, case_count, count, n
population	Yes	Population denominator used to calculate prevalence.	population, pop, denominator

The application also requires a configured GeoPackage for the active country. Administrators map each geographic level to a GeoPackage layer, a spatial identifier, and a display-name column in config.json.

Identifier matching. Geographic identifiers in the CSV must match either the configured map IDs or labels after standardisation and zero padding. Leading zeros are significant.

3. Configure and run

1. Upload one or more CSV files.
2. Use Manage files to activate, deactivate, remove, or clear files.
3. Select a Country when several are configured.
4. Choose the Geographic Level.
5. Choose Single Week or Week Range.
6. Select the Year and epidemiological week or range.
7. Select Run Analysis.

Control	Current behavior
Manage files	All active files are combined. Deactivating a file removes it from the current analysis without deleting the original upload.
Geographic Level	Selects the administrative level and its configured GeoPackage layer.
Time Mode - Single Week	Uses one epidemiological week.
Time Mode - Week Range	Aggregates cases over the selected inclusive range.
Year	Limits the analysis to one epidemiological year.
Run Analysis	Prepares the data, checks spatial configuration, loads geometry, joins records, computes LISA, and refreshes maps and tables.

For week ranges, cases are summed across weeks. Population is treated as a static denominator: detailed geographic records are deduplicated before population is rolled up to the selected level, preventing repeated weekly population values from being added multiple times.

4. Review the outputs

Global Moran's I banner

The banner above the maps reports Global Moran's I and its p-value. A significant positive value suggests nearby areas tend to have similar prevalence; a value near zero suggests little overall spatial pattern; a negative value suggests neighbouring areas tend to differ.

Prevalence Map

The choropleth displays the configured prevalence measure, normally cases per 100,000 people. The legend remains visible. Popups identify the area, map identifier, prevalence, LISA category, and p-value when available.

LISA Cluster Map

Category	Meaning
High-High	An area with high prevalence surrounded by areas with high prevalence; a local hotspot pattern.
Low-Low	A low-prevalence area surrounded by low-prevalence areas.
High-Low	A high-prevalence area surrounded by low-prevalence areas; a spatial outlier.
Low-High	A low-prevalence area surrounded by high-prevalence areas; an inverse outlier.
Not Significant	The local pattern did not meet the current p <= 0.05 significance rule.

Results Table

The results table contains identifiers, names, cases, population, weeks included, prevalence, LISA category, and p-value as available. Use Search by to select a text or numeric column and filter the display.

Prepared Data

Shows the aggregated dataset before geometry display, plus the data-preparation log. It records time filtering, case aggregation, population deduplication, prevalence calculation, and the number of prepared spatial units.

Display settings. The Help tab can increase interface font size and change table decimal places. Downloaded CSV

values retain full precision.

5. Interpret results responsibly

- Global Moran's I summarizes the entire selected map; it does not identify which individual areas form clusters.
- LISA provides local categories, but multiple testing and sparse data should be considered during interpretation.
- A significant cluster may reflect surveillance intensity, population structure, access to services, or reporting practices as well as underlying risk.
- Compare maps across justified time windows rather than choosing a window only because it produces a desired pattern.

Before acting on a hotspot or outlier, check the numerator, population denominator, identifier match, selected weeks, and whether neighbouring polygons are epidemiologically meaningful for the question.

6. Downloads

Download	Contents
Download Results CSV	Tabular geospatial results for all matched units.
Download Prepared Data CSV	Aggregated input used for the spatial join and prevalence calculation.
Map export, where enabled	Image of the current interactive map. Availability depends on the screenshot packages installed in the deployment.

On-screen filters and decimal formatting do not change the underlying downloaded result values.

7. Troubleshooting

Problem	What to check
Geographic data are not available	The configured raw_data/<file_name>.gpkg file is missing.
Layer or column pre-flight error	Check layer_name, spatial_id, name_col or label_col in config.json.
Spatial Join Failure	CSV codes do not match map IDs or labels; check leading zeros, level selection, and code length.
No data available	The active files may not contain the selected year or weeks.
No Global Moran banner	Fewer than four matched units, all values missing, zero variance, or an invalid neighbour structure.
LISA values are blank	The same sparse/constant-data conditions can prevent local analysis.
Population error	Population is missing, zero, invalid, or mapped under an unrecognised alias.

Escalation package. Include the selected country, level, year, week window, active file names, preparation log, pre-flight message, and a small non-sensitive sample of the geographic IDs.

8. Glossary

Term	Meaning
Prevalence per 100,000	Cases divided by population, multiplied by 100,000.
Global Moran's I	Whole-map measure of spatial autocorrelation.
Local Moran's I / LISA	Local statistic used to classify clusters and spatial outliers.
Spatial weights	Row-standardised neighbour relationships derived from polygon contiguity.
GeoPackage	Spatial database file containing configured polygon layers.
Spatial join	Match between prepared analysis_unit_id values and configured map IDs or labels.